Task 5 — Implementation and Testing Report

Group 7 · Software Engineering for CSAI · 6 November 2026

Overview

We defined **5 unit tests (U1–U5)** and **3 integration tests (I1–I3)**. Each one traces to a requirement in our SRS, and all 8 pass. The tests are written in **pytest** (`tests/`) and run automatically in **GitHub Actions** on every push and pull request. On 6 Nov 2026, `main` ran 171 test functions, all passing, in 54 s. That count includes the parametrised cases and extra regression tests.

Table 1 — Test cases, test data and results

ID	Req.	Test case	Test data (input)	Expected result	Result
Unit tests — one function each; trained models replaced by fakes with known outputs					
U1	FR-1	`validate_interest()` accepts a valid interest and refuses an empty one	"I want to work in data science"; " machine learning "; ""; " "; None; 500 × "a"	Valid text and exactly 500 chars accepted; whitespace trimmed. Empty, blank and None refused with the SRS error message (no exception)	Pass
U2	FR-2, NFR-3	`validate_study_context()` checks language, hours, weeks; labels missing fields	Hours 1, 40 vs 0, 41; weeks 1, 52 vs 0, 53; non-text language; blank and NaN fields	Limits accepted. Out of range refused with "Hours per week must be a number from 1 to 40" (same for weeks). Missing fields shown as "not given"	Pass
U3	FR-13, AIR-3	`passes()` compares a score with its threshold	Skill threshold 0.35; scores 0.30, 0.35, 0.50	False, True, True (a score exactly at the threshold is kept)	Pass
U4	FR-14, AIR-1	`classify()` returns Model 1's top category and confidence	Fake model with fixed probabilities; title "Python" + intro "Learn the basics"; empty title and intro	("Computer Science", 0.70) as plain `str`/`float`; empty input raises `ValueError` before the model is called	Pass
U5	FR-15, AIR-2	`rank()` / `match()` return the best ESCO skills, sorted	Fake keyword encoder over a 3-skill index; query "python programming"; k = 2, 5, 50	At most k (≤ 5 for `match`) results, highest score first; "Python (computer programming)" ranked first	Pass
Integration tests — real components together, no mocks					
I1	FR-13, FR-14	Real Model 1 + real `thresholds.json` decide what the user sees	Vague course "General introduction"; clear course "Python for Data Science"; category threshold 0.20	Vague course (confidence ≈ 0.15) shown as "uncategorised"; clear course (≈ 0.69) kept as "Data Science"	Pass
I2	FR-13, FR-15, AIR-2	Real encoder + ESCO index + thresholds find skills by meaning	25 ESCO skills (seed 42), each queried by an alt label that is first removed from the index; nonsense "qzxv jkwp"	≥ 85% found in the top 5; each accepted match has an ESCO ID and a score in [0, 1] at or above 0.45; nonsense gives no accepted match	Pass
I3	FR-4, FR-13–15, AIR-8	Batch catalogue preparation with real Model 1 and Model 2	10 real Coursera courses: 5 blank categories, 1 non-canonical ("Finance"), 1 localised ("Negocios"), 1 without skills, 1 empty intro	Every course gets one of the 11 categories; each distinct skill gets its top-5 ESCO matches; output feeds retrieval; each lookup < 500 ms	Pass

Why we chose these tests

- **Unit tests isolate one function.** U1–U3 call pure functions with fixed inputs. U4 and U5 replace the trained models with small fakes whose outputs we know in advance. So a failure points to our code, not to model behaviour.
- **Integration tests use the real components.** I1–I3 load the real Model 1 artefact, the real sentence encoder, the ESCO index and the threshold configuration, with no mocks. They check that the parts give correct results together.
- **Boundaries and edge cases.** We test values exactly on the limits: 500 characters, 1/40 hours, 1/52 weeks, and a score equal to the threshold. We also test messy real-data cases: blank, localised and non-canonical labels, missing skills and empty intros.
- **The riskiest requirements.** The tests cover both AI models (FR-14, FR-15), the threshold that withholds weak AI output (FR-13, AIR-3) and the 500 ms limit (AIR-8). A silent error in any of these would mislead the user.